Technical guide

Forwarding Booking Module Mapping Guide

Universal XML

31 October 2025

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Date	Version	Description of Change	Reference
04 Jun 2025	1.01	Added GreenhouseGasCollection and GreenhouseGas	WI00885977
31 Oct 2025	1.02	Clarity added to 1.1 Data Source/Target	CS02116050

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1. Introduction

The Forwarding Bookings module allow freight forwarders and NVOCC to take bookings (quick bookings or bookings with quotes) from their respective customers for future consolidation and shipment. The module supports both, full container loads and loose freight for all models of transport providing tracking, documentation, and possible consolidation as part of its interface.

1.1 DataSource / DataTarget Type – ForwardingBooking

A universal shipment targeted for import into this module becomes a forwarding booking. If a matching booking is found in the recipient system, this booking will be updated.

It is possible to receive a universal shipment XML with Target Type ForwardingBooking into this module, but if a booking was already converted to a shipment/consolidated, the XML file with Target Type Forwarding Booking will be rejected.

If a universal shipment XML contains consol information and shipment details matching an existing booking, CargoWise will create or find a matching consolidation and then convert the matching booking to a shipment and attach this shipment to the consolidation.

Refer to our Forwarding Booking learning units on MyAccount web portal for more information.

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2. Universal events

2.1 Data export

When you export a universal event from forwarding booking, the following information will be extracted and included in ContextCollection of the XML.

Context Elements

Context type	Value mapped from	Notes
HAWBNumber/HBOLNumber	**Additional Details** tab, **House Bill Number** field	HAWBNumber for air and HBOLNumber for other transport modes
HAWBOriginIATAAirportCode	The IATA code of the origin UNLOCO	Air only
HAWBDestinationIATAAirportCode	The IATA code of the destination UNLOCO	Air only
HBOLOriginUNLOCO	**Details** tab, **Origin** field	
HBOLDestinationUNLOCO	**Details** tab, **Destination** field	
ShippersReference	**Details** tab **Shipper's Ref** field	

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Context type	Value mapped from	Notes
OrderNumber	**Details > Job Management Links > More** tab, **Order References** field	
InterimReceipt	**Additional Details** tab, **Interim Receipt** field	
Reference Numbers	**Additional Details** tab, **Reference Numbers** field	Each type of reference number and description serve as their own context type and description
CFSReference	**Additional Details** tab, **CFS Reference** field	

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2.2 Data import

2.2.1 Matching and special update logic

- **Job Number Matching** – this is the first matching strategy the system will use when importing a universal event XML. You can see from the XML snippet below, DataContext contains DataTargetCollection describing the importing target. A DataTarget contains both type and key to indicate the type of the target this event is importing to (in this case, ForwardingBooking) and the job number of the parent. The system will look for the data target and try to load the event parent with this information.

```
<Event>
  <DataContext>
    <DataTargetCollection>
      <DataTarget>
        <Type>ForwardingBooking</Type>
        <Key>S00001003</Key>
      </DataTarget>
    </DataTargetCollection>zz
```

- **Combination Key Match** – if key was not defined in XML or job with this job number is not found, a combination key match will be conducted to find the best matching ForwardingBooking. The elements of ContextCollection described in above Data Export section will be used to match. The detailed process as described below.
- **Reference Match Count** – during reference match, HAWBNumber/HBOLNumber, ShippersReference, InterimReceipt, CFSReference and Reference numbers inside ContextCollection will be used to match information in booking. The system will find the booking that matches the most of these fields.
- **Reference Type Weighting** - if multiple bookings with the same number of fields matched, the system will try to match according to the order of the Context Elements described above. The order of the Context Elements in the table above represents the weight or priority of the element. The element at a higher order in the table will have higher weight. The system will pick the one that matches fields with bigger weight.
- **Context Fallback Matching** - if there are still multiple results exist, the system will use OrderNumber, HBOLOriginUNLOCO and HBOLDestinationUNLOCO to match booking as a fallback. During fallback match, the system will first try to find booking with at least one fallback criteria matched, if multiple results still exist, the system will try to find the one with all fallback criteria matched.
- **Most Recent Date Match** – where there are multiple matches found with the same score or weighting, it will pick the booking with more recently created date.

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3. Universal shipment

3.1 Data export

Unlike when exporting a forwarding shipment, when a forwarding booking is exported, the top-level universal shipment is populated direct from the booking. There is no consolidation at the booking stage, so everything goes on the top-level universal shipment.

A booking-style universal shipment can still be imported in the Forwarding Shipment module. Information like containers and transport legs that are stored against the consol in the context of a forwarding shipment will be stored in the appropriate place upon importing.

Collections are populated in the following manner:

- **ContainerCollection** - populated from the **Containers** grid on the **Details** tab of the booking.
- **PackingLineCollection** - populated from the **Loose Cargo** grid at the bottom of the Booking form. This only contains loose packages in the case of the Booking module with no container references.
- **OrganizationAddressCollection** - looking at the **Main** tab of the forwarding Booking, the consignor, consignee and local client organizations along with the consignor pickup address and consignee delivery address from the **Additional Details** tab are populated here.
- **NoteCollection** – all notes from the **Notes** tab are included.
- **AdditionalReferenceCollection** – all additional references linked to the forwarding booking are mapped. Populated from **Reference Numbers** grid located at the bottom right of **Additional Details** section on booking.
- **DateCollection** - the dates exported from the forwarding shipment at the time of publication of this document were **Booking Confirmed**, **Received**, **Departure**, **Arrival**, and **Client Requested ETA** fields.
- **EntryNumberCollection –** all entry numbers linked to the forwarding booking are mapped. Entry number is populated from **Brokerage Details** section on the booking.

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3.2 Data import

3.2.1 Matching and special update logic

- **Context Key Matching** – this is the first matching strategy the system will use when importing a universal shipment XML. As you can see from the XML snippet below, DataContext contains DataTargetCollection describing the importing target. A DataTarget contains both type and key to indicate the target booking to be imported. The system will use the key in DataTarget, which is the JobNumber of the booking, to try to load the booking.

```
<UniversalShipment xmlns="http://www.cargowise.com/Schemas/Universal/2011/11"
version="1.1">
  <Shipmen>
    <DataContext>
      <DataTargetCollection>
        <DataTarget>
          <Type>ForwardingBooking</Type>
        </DataTarget>
      </DataTargetCollection>
```

- **Combined Reference and Party ID Matching** – if Registry eServices - Universal XML-Use Combined Reference and Party ID Matching is enabled; the system will try to match with reference and party ID. The system has a list of pre-defined criteria specifying which reference and organization will be used for matching.

Each criteria has three score, the Match score for matching both reference and party, the Partial score for matching only reference, the Conflict score is for the case where reference is provided but it is different from the given job. All the matched score will be sum to a total and compared in order to get the best match.

All the criteria of forwarding booking

Reference element	Party	Match	Partial	Conflict
BookingConfirmationReference	Consignor	90	10	0
BookingConfirmationReference	Consignee	90	10	0
InterimReceiptNumber	Consignor	90	10	0
WayBillNumber	SendingForwarder	180	30	-180

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Note: the criteria to take effect, the **Reference** field must be longer than six characters. Unmatched organization will also be taken into considerations if enabled in the registry.

- **Combination Key Match** – the combination key match of for booking is similar to how it imports event, except for in here, we do not use HAWBOriginIATAAirportCode and HAWBDestinationIATAAirportCode to match.
- **Module Specific Business Rules** – each module can have their own specific logic for how they want to find a match based on elements they have. Currently, no module specific business rule will be used when importing a forwarding booking.

  However, for RelatedShipmentCollection inside booking (which contains order information attached to the shipment), pre-defined module specific business rules will be used to find the best match. The system will search for a bookings that matches both buyer (Order By) and order number. Then it will find the first order with the same order number split.

3.3 Collection mappings

Collections are mapped in the following manner:

- **RelatedShipmentCollection** – used to store order information that is attached to the booking. Each related shipment represents one of the order that is attached to the booking. A context key matching will be used to find a match based on DataTarget key and value. The context value will be a combination of order number, order number split and buyer code. If no result is found, it will then try to match based on module specific business rules (as described under 3.2.1 Matching and Special Update Logic).
- **Local Processing > OrderNumberCollection** – matched on order reference. It will also remove any order reference that was not in XML.
- **Local Processing > AdditionalServiceCollection** – matched on service code.
- **AdditionalReferenceCollection** – matched on type if the type of reference need to be unique in the collection, otherwise match on ReferenceNumber as well,
- **ContainerCollection** – match on container number, fallback to match on container type.
- **DateCollection** – matched based on type of date. The following is all the date types that will be imported to ForwardingBooking and their target location on GUI.
- **EntryNumberCollection** – matched on type and CountryOfIssue. EntryNumbers are always brought in ignoring the EntryIsSystemGenerated flag. System generated entry numbers will not be updated, if they already exist in CargoWise and are read only. If there is no pre-existing system generated entry number, a new entry number row will be added with the IsSytemGenerated flag set to false.
- **NoteCollection** – matched on note description and context (module, direction, and freight). If all the context values are set to ALL, it can be matched with either ALL context or empty context.
- **OrganizationAddressCollection** – matched on Organization Code and Address Short Code, fallback to match by address similarity, if still not found and no other address information exist, match by custom registration number, and finally fallback to unmatched organization

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in the registry if enabled.

- **PackingLineCollection** – if there is only one packing line on both universal XML and target booking, then match with UNDG on the packing line (see below for details). The UNDG matching is done though substance code, if no code or the codes are not matched, fallback to match IMO Class.

3.4 Element mappings

3.4.1 Shipment

Element name (XML)	Value mapped from	Notes
ActualChargeable	**Details** tab, **Goods Details** section, **Chargeable** field	
AdditionalTerms	**Details** tab, **Add. Terms** field	
AWBServiceLevel	**Additional Details** tab **Sailing Summary** section **Carrier Service Level** field	Shows only when the **Direct** box is ticked.
BookingConfirmationReference	**Details** tab, **Shipper's Ref** field	
CFSReference	**Additional Details** tab, **CFS Reference** field	
ContainerCount	Calculated from the total number of containers on booking	
ContainerMode	**Details** tab, **Mode** field	LRO, LRA will be mapped to LCL, FRO, FRA will be mapped

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Element name (XML)	Value mapped from	Notes
		to FCL, COU will be mapped to OBC
FreightRate	**Details > Monetary Values** tab, **Sport Rate** section	
FreightRateCurrency	**Details** tab, **Monetary Values** section, **Sport Rate** field	
GoodsDescription	**Details** tab, **Description** field	
GoodsValue	**Details** tab, **Monetary Values** section, **Goods Val** field	
GoodsValueCurrency	**Details** tab, **Monetary Values** section, **Goods Val** field	
GreenhouseGasEmission	**Booking > CO2e (Kg)** field	<CO2e>, <CO2eDistanceInKm>, <CO2ePerTEU>, <CO2ePerTonne>, <CO2eStatus>, <GreenhouseGasCollection>, <GreenhouseGas>
HBLAWBChargesDisplay	**Additional Details** tab, **Charges Apply** field	Show when NVOCC Display box is ticked

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Element name (XML)	Value mapped from	Notes
InsuranceValue	**Details** tab, **Monetary Values** section, **Ins Value** field	
InsuranceValueCurrency	**Details tab > Monetary Values > Ins Value**	
InterimReceiptNumber	**Additional Details** tab, **Interim Receipt** field	
IsDirectBooking	**Additional Details** tab, **Sailing Summary** section, **Direct** field	
LloydsIMO	**Additional Details** tab, **Sailing Summary** section, **View Sailings** tab, **Vessel** section, **Lloyds/IMO Number** field	
OuterPacks	**Details** tab, **Goods Detail** section, **Outers Value** field	
OuterPacksPackageType	**Details** tab, **Goods Detail** section, **Outers Value** field	
ShipmentIncoTerm	**Details** tab, **Incoterm** field	Show when **NVOCC Display** box is ticked
PortOfDestination	**Details** tab, **Destination** field **Additional Details** tab, **Dest** field	

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Element name (XML)	Value mapped from	Notes
PortOfDischarge	**Additional Details** tab, **Disch**. field	
PortOfLoading	**Additional Details** tab, **Load** field	
PortOfOrigin	**Details** tab, **Origin** field **Additional Details** tab, **Origin** field	
ReleaseType	**Additional Details** tab, **Release Type** field	Show when **NVOCC Display** box is ticked
ServiceLevel	**Details** tab, **Service Level** field	
ShippedOnBoard	**Additional Details** tab, **On Board** field	Show when **NVOCC Display** box is ticked
TEU	**Booking > Containers** grid **> Container Type > TEU Count** field	Internal Use only – This tag also contains sub tag ContainerEmptyWeightPerTEU taken from **Container Type** ref file > **Tare Weight** field
TotalVolume	**Details** tab, **Goods Detail** section, **Volume** field	
TotalVolumeUnit	**Details** tab, **Goods Detail** section, **Volume** field	

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Element name (XML)	Value mapped from	Notes
TotalWeight	**Details** tab, **Goods Detail** section, **Weight** field	
TotalWeightUnit	**Details** tab, **Goods Detail** section, **Weight** field	
TransportMode	**Details** tab, **Mode** (converted) field	
VesselName	**Additional Details** tab, **Sailing Summary** section, **Vessel** field	If vessel name is empty, fall back to the Lloyd's/IMO Number to search and match the vessel with Lloyd's/IMO Number from the database.
VoyageFlightNo	**Additional Details** tab, **Sailing Summary** section, **Voyage No**. field	
WayBillNumber	**Additional Details** tab, **Sailing Summary** section, **Master Bill Number** field	
WayBillType	Hard coded to "HWB"	
DeliveryRequiredBy	**Details** tab, **Est Delivery** section, **Required By** field	
EstimatedDelivery	**Details** tab, **Est Delivery** field	
EstimatedPickup	**Details** tab, **Est Pickup** field	.

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Element name (XML)	Value mapped from	Notes
FCLDeliveryEquipmentNeeded	**Details** tab, **Goods Detail** section, **Dlv. Drop** field	
FCLPickupEquipmentNeeded	**Details** tab, **Goods Detail** section, **Pic. Drop** field	
InsuranceRequired	**Details** tab, **Monetary Values** section, **Insurance Required** field	
PickupRequiredBy	**Details** tab, **Est Pickup** section, **Required By** field	

3.4.2 LocalProcessing / AdditionalServiceCollection

Element name (XML)	Value mapped from
ServiceCode	**Additional Details** tab, **Service** grid, **Type** field
Booked	**Additional Details** tab, **Service** grid, **Date Booked** field
Completed	**Additional Details** tab, **Service** grid, **Completed** field
Contractor	**Additional Details** tab, **Service** grid, **Contractor** field

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3.4.3 OrderNumberCollection

Element name (XML)	Value mapped from
Sequence	Set by system
OrderReference	**Details** tab, **Job Management Links** section, **Order Refs** field

3.4.4 AdditionalReferenceCollection

Element name (XML)	Value mapped from
Type	**Additional Details** tab, **Reference Numbers** grid, **Type** field
CountryOfIssue	**Additional Details** tab, **Reference Numbers** grid, **Country/Region of Issue** field
ReferenceNumber	**Additional Details** tab, **Reference Numbers** grid, **Reference Number** field
ContextInformation	**Additional Details** tab, **Reference Numbers** grid, **Additional Information** field
IssueDate	**Additional Details** tab, **Reference Numbers** grid, **Issue Date** field

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3.4.5 ContainerCollection

Element name (XML)	Value mapped from
ContainerNumber	**Details** tab, **Containers** grid, **Container #** field
ContainerCount	**Details** tab, **Containers** grid, **Count** field
ContainerType	**Details** tab, **Containers** grid, **Type** field
FCL_LCL_AIR	**Details** tab, **Containers** grid, **Mode** field
Commodity	**Details** tab, **Containers** grid, **Commodity** field
ReleaseNum	**Details** tab, **Containers** grid, **Release** field
DepartureEstimatedPickup	**Details** tab, **Containers** grid, **Full Pickup** field
DepartureSlotDateTime	**Details** tab, **Containers** grid, **Dep. Slot Date** field
DepartureSlotReference	**Details** tab, **Containers** grid, **Dep. Slot Ref.** field
EmptyRequired	**Details** tab, **Containers** grid, **Empty Required** field
TempRecorderSerialNo	**Details** tab, **Containers** grid, **Temp Recorder Serial Num** field
SetPointTemp	**Details** tab, **Containers** grid, **Set Point Temp**. field
AirVentFlow	**Details** tab, **Containers** grid, **Air Vent** field

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3.4.6 DateCollection

Elements in the DateCollection will be mapped to different date control on booking.

Element name (XML)	Value mapped from
BookingConfirmed	**Additional Details** tab, **Booked Date** field
Received	**Additional Details** tab, **Warehouse Rec**. field
Departure	**Additional Details** tab, **ETD** field
Arrival	**Additional Details** tab, **ETA** field
ClientRequestedETA	**Additional Details** tab, **Client Req. ETA** field

3.4.7 Notecollection

Element name (XML)	Value mapped from
Description	**Notes** tab, **Description** field
IsCustomDescription	**Notes** tab, **Custom** field
NoteText	**Notes** tab, **Note** field
NoteContext	**Notes** tab, **Module/Direction/Freight** field
Visibility	**Notes** tab, **Visibility** field
ContainerCount	**Notes** tab, **Company** field

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3.4.8 OrganizationAddressCollection

Element name (XML)	Value mapped from
ConsignorDocumentaryAddress	**Details** tab, **Consignor** field
ConsignorPickupDeliveryAddress	**Additional Details** tab, **Pickup Address** field
ConsigneeDocumentaryAddress	**Details** tab, **Consignee** field
ConsigneePickupDeliveryAddress	**Additional Details** tab, **Delivery Address** field
PickupAgent	**Additional Details** tab, **Pickup Agent** field
ControllingAgent	**Additional Details** tab, **Controlling Agent** field
ControllingCustomer	**Additional Details** tab, **Controlling Customer** field
ExportBroker	**Additional Details** tab, **Export Broker** field
DepartureCFSAddress	**Additional Details** tab, **CFS** field
DeliveryAgent	**Additional Details** tab, **Delivery Agent** field
PickupLocalCartage	**Additional Details** tab, **Port Transport** field
SendersLocalClient	**Details** tab, **Client** field
ShippingLineAddress	**Additional Details** tab, **Carrier** field
Creditor	**Additional Details** tab, Creditor

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3.4.9 PackingLineCollection

Element name (XML)	Value mapped from
Commodity	**Details** tab, **Loose Cargo** grid, **Commodity** field
EndItemNo	**Details** tab, **Loose Cargo** grid, **Items** field
GoodsDescription	**Details** tab, **Loose Cargo** grid, **Goods Description** field
HarmonisedCode	**Details** tab, **Loose Cargo** grid, **HarmonisedCode** field
Height	**Details** tab, **Loose Cargo** grid, **Height** field
ItemNo	**Details** tab, **Loose Cargo** grid, **Item No** field
Length	**Details** tab, **Loose Cargo** grid, **Length** field
LengthUnit	**Details** tab, **Loose Cargo** grid, **UD** field
LinePrice	**Details** tab, **Loose Cargo** grid, **Line Price** field
MarksAndNos	**Details** tab, **Loose Cargo** grid, **Marks & Numbers** field
PackQty	**Details** tab, **Loose Cargo** grid, **Packs** field
PackType	**Details** tab, **Loose Cargo** grid, **Packages** field
ReferenceNumber	**Details** tab, **Loose Cargo** grid, **Ref Number** field
Volume	**Details** tab, **Loose Cargo** grid, **Volume** field

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Element name (XML)	Value mapped from
VolumeUnit	**Details** tab, **Loose Cargo** grid, **UV** field
Weight	**Details** tab, **Loose Cargo** grid, **Weight** field
WeightUnit	**Details** tab, **Loose Cargo** grid, **UW** field
Width	**Details** tab, **Loose Cargo** grid, **Width** field

3.4.10 PackingLineCollection / UNDGCollectionCollection

Element name (XML)	Value mapped from
Contact	**Details** tab, **Loose Cargo** grid, **DG Contact** field
FlashPoint	**Details** tab, **Loose Cargo** grid, **Flash Point** field
IMOClass	**Details** tab, **Loose Cargo** grid, **DG Class** field
MarinePollutant	**Details** tab, **Loose Cargo** grid, **Marine Pollutant** field
TechnicalName	**Details** tab, **Loose Cargo** grid, **Technical Name** field
UNDGCode	**Details** tab, **Loose Cargo** grid, **DG Subs** field
Volume	**Details** tab, **Loose Cargo** grid, **DG Volume** field
VolumUQ	**Details** tab, **Loose Cargo** grid, **DG UV** field

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Element name (XML)	Value mapped from
Weight	**Details** tab, **Loose Cargo** grid, **DG Weight** field
WeightUQ	**Details** tab, **Loose Cargo** grid, **DG UW** field

4. Sample files

Sample files are compiled into a .zip file format which can be downloaded and extracted. Listed below is the name of the sample files provided for this guide

Refer to the Forwarding Module Mapping Guides under **WiseTech Academy > CargoWise > Technical Guides**.

4.1 Universal shipment

- **UniversalShipment_ForwardingBooking_DataTarget_SEA.xml** – this UniversalShipment XML has a DataTarget > Key, it will find and update the existing SEA Forwarding Booking S54632311 in CargoWise.
- **UniversalShipment_ForwardingBooking_DataTarget_AIR.xml** – this UniversalShipment XML will create a Forwarding Booking with AIR transport mode in CargoWise.

4.2 Universal event

- **UniversalEvent_BKD_HBOLNumber_Context.xml** – this UniversalEvent does not have a DataTarget, it will use the HBOLNumber to match and link to a forwarding booking while importing into CargoWise. It will link event, if a booking exists with HBOLNumber S546323116.
- **UniversalEvent_BKC_ForwardingBooking_DataTarget.xml** – this UniversalEvent will use the DataTarget to match the forwarding booking while importing to CargoWise. It will link event, if a booking exists with booking number S0001000.

Support: Press the **F1** key from anywhere within CargoWise to raise an eRequest.

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